

Fundamentals of Electric Circuits

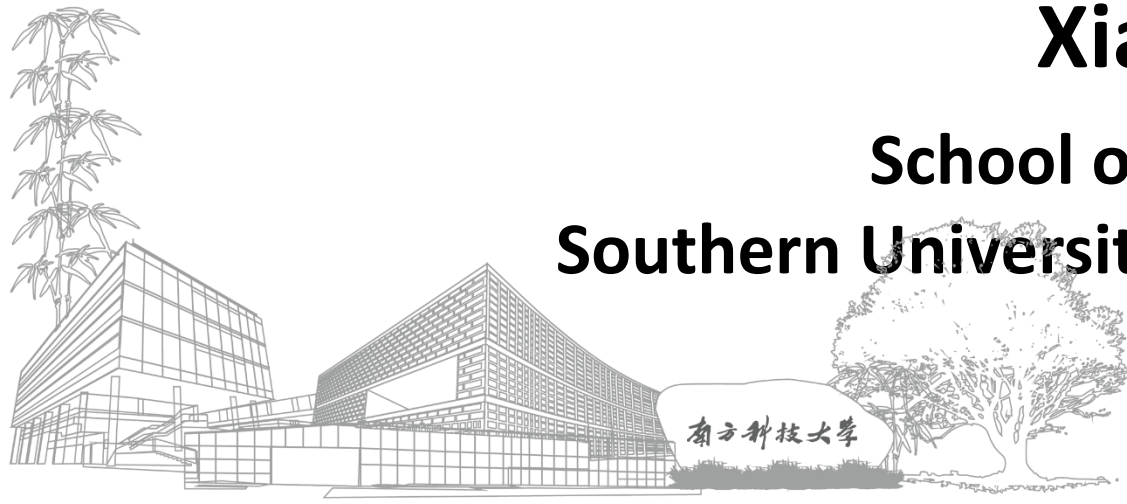
电路基础

Lecture12: Magnetically Coupled Circuits

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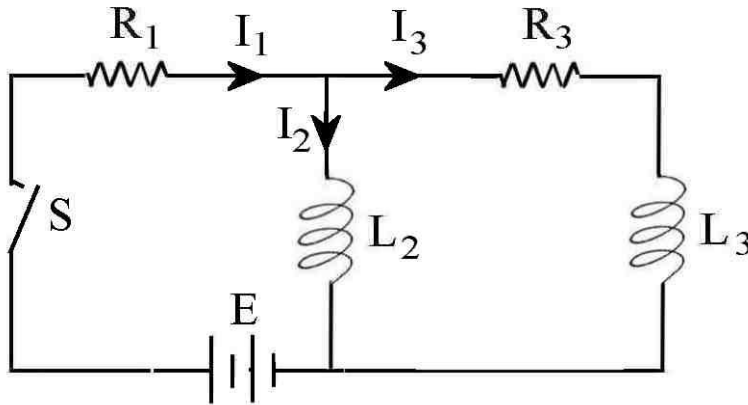


Outline

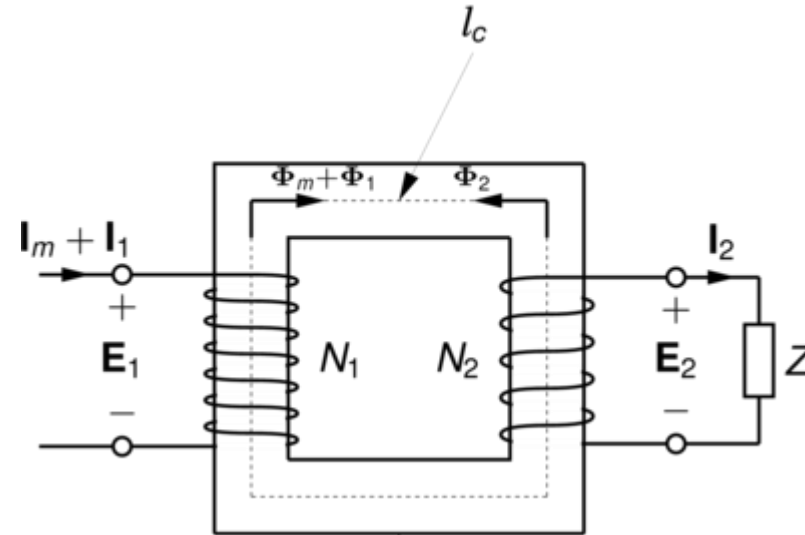
- **Introduction**
- **Mutual inductance**
- **Linear transformers**
- **Ideal transformers**

Introduction

- All circuits we have learned so far can all be regarded as **conductively coupled (电耦合)**, i.e., one branch in the circuit affects other branches through current conduction
- In this lecture, we will learn a new type of electric element that takes effect through **magnetic field coupling (磁场耦合)**



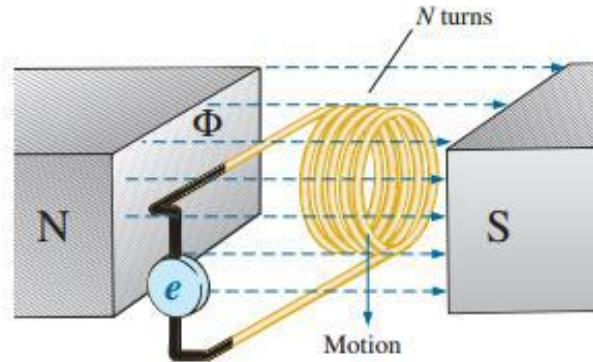
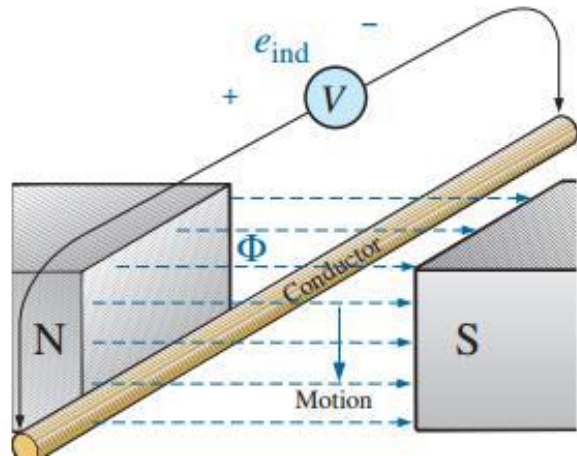
Conductively Coupled



Magnetically Coupled

Faraday's law of Electromagnetic Induction

- Faraday's law of electromagnetic induction (法拉第电磁感应定律): Apart from the current generated from charges, current can also be generated from moving wire in a fixed magnetic field or a fixed coil in a time-varying magnetic field
- The magnitude of the electromagnetic field induced in a circuit is proportional to the rate of change with time t of the magnetic flux ϕ (磁通量), that cuts across the circuit



$$e = N \frac{d\phi}{dt}$$

Self-Inductance

- According to Faraday's law, the voltage v induced in the coil is proportional to the number of turns N and the time rate of change of the magnetic flux, that is

$$v = N \frac{d\phi}{dt} = N \frac{d\phi}{di} \frac{di}{dt} = L \frac{di}{dt}$$

- Self-inductance (自感) relates the voltage induced in a coil with the rate of a time-varying current in the same coil. The inductance L of the inductor is

$$L = N \frac{d\phi}{di}$$

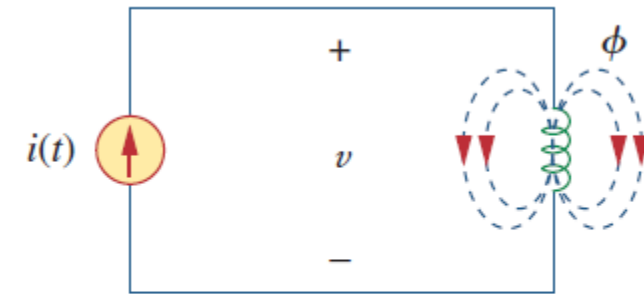


Figure 13.1

Magnetic flux produced by a single coil with N turns.

Mutual Inductance (I)

- When two inductors are closely placed, the magnetic flux generated by the first coil, ϕ_1 , would consist of two parts: 1) those only linked with itself, ϕ_{11} , and 2) those linked with the second coil, ϕ_{12} , as illustrated in the figure

$$\phi_1 = \phi_{11} + \phi_{12}$$

- Although the two coils are physically separated, they are said to be magnetically coupled

$$v_1 = N_1 \frac{d\phi_1}{di_1} \frac{di_1}{dt} = L_1 \frac{di_1}{dt}$$

$$v_2 = N_2 \frac{d\phi_{12}}{di_1} \frac{di_1}{dt} = M_{21} \frac{di_1}{dt} \Rightarrow M_{21} = N_2 \frac{d\phi_{12}}{di_1}$$

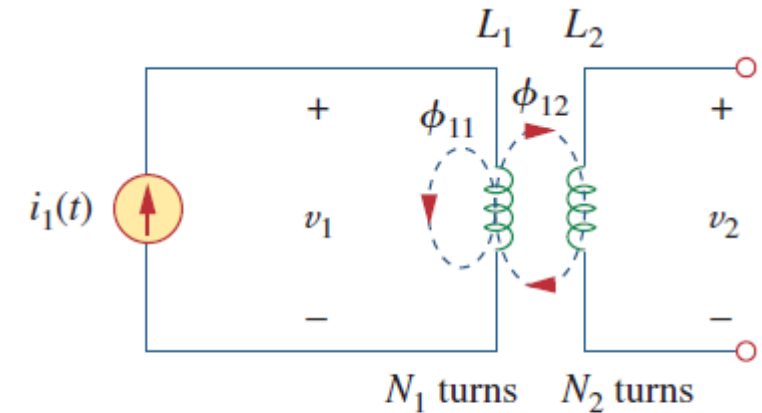


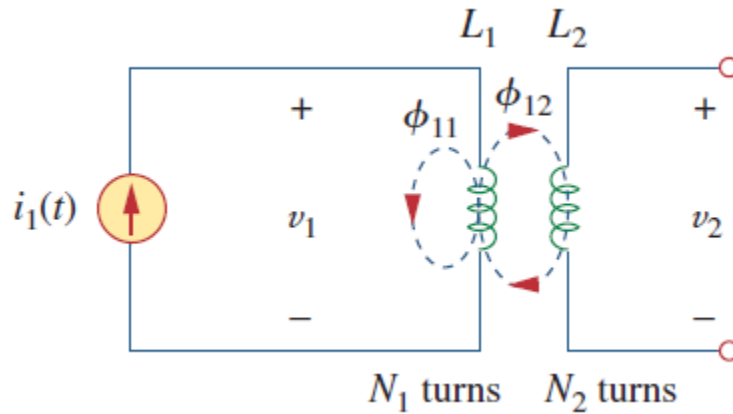
Figure 13.2

Mutual inductance M_{21} of coil 2 with respect to coil 1.

- Where M_{21} is called **mutual inductance** (互感)
- M_{21} is always positive and measured in Henries (H)

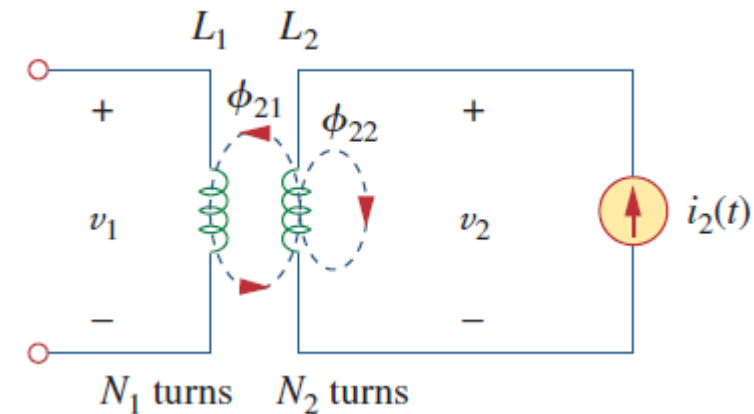
Mutual Inductance (II)

- Mutual inductance is the ability of one inductor to induce a voltage across a neighboring inductor, measured in henrys (H)



Coil 1 $\rightarrow$ Coil 2

$$v_2 = M_{21} \frac{di_1}{dt} \quad M_{21} = N_2 \frac{d\phi_{12}}{di_1}$$



Coil 2 $\rightarrow$ Coil 1

$$v_1 = M_{12} \frac{di_2}{dt} \quad M_{12} = N_1 \frac{d\phi_{21}}{di_2}$$

$$M_{12} = M_{21} = M$$

Mutual Inductance: Dot Convention (I)

- The choice of the correct polarity for $M di/dt$ is made by examining the orientation or particular way in which both coils are physically wound → Inconvenient to show the construction details of coils on a circuit schematic
- If a current **enters** the **dotted terminal** (同名端) of one coil, the reference polarity of the mutual voltage in the second coils is **positive** at the dotted terminal of the second coil

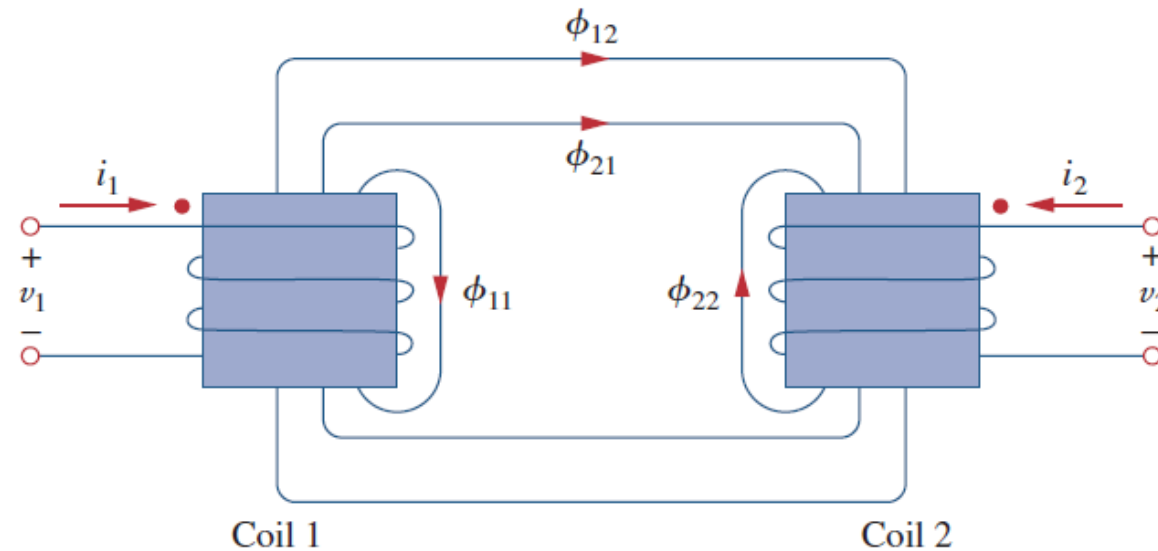
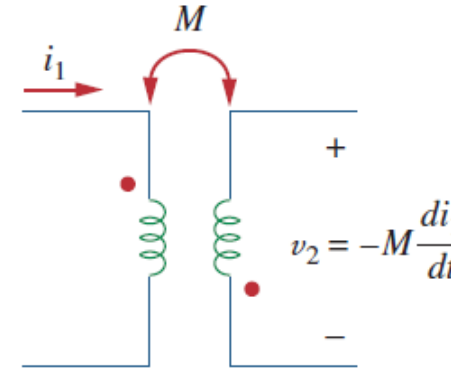
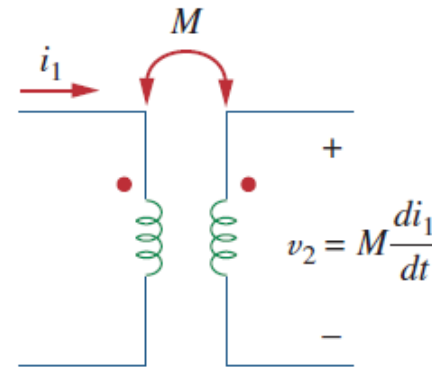


Figure 13.4

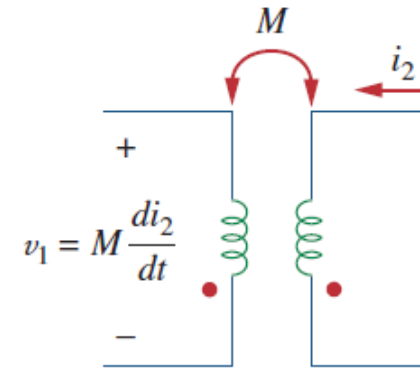
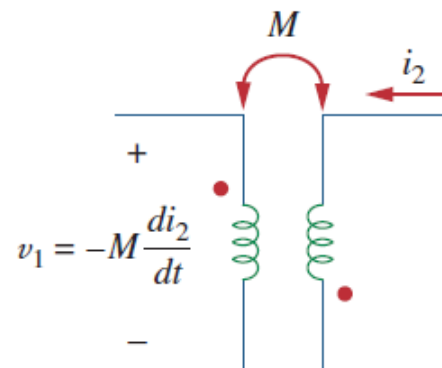
Illustration of the dot convention.

Mutual Inductance: Dot Convention (II)

- If a current **enters** the **dotted terminal** of one coil, the reference polarity of the mutual voltage in the second coil is **positive** at the **dotted terminal** of the second coil

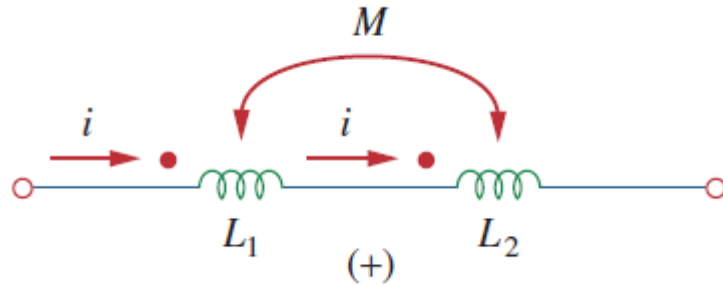


- If a current **leaves** the **dotted terminal** of one coil, the reference polarity of the mutual voltage in the second coil is **negative** at the **dotted terminal** of the second coil



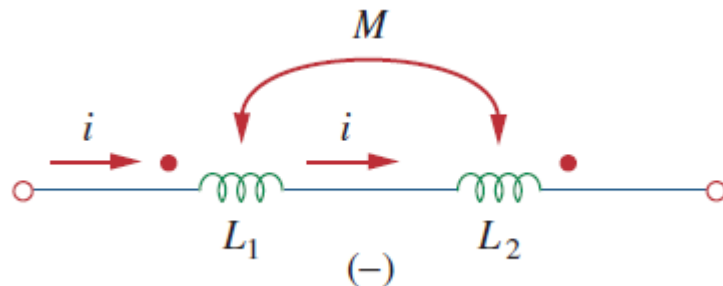
Coils in Series

- The coupled coils can be connected in series in two different ways
 - Series aiding connection



$$L = L_1 + L_2 + 2M \quad (\text{Series-aiding connection})$$

- Series opposing connection



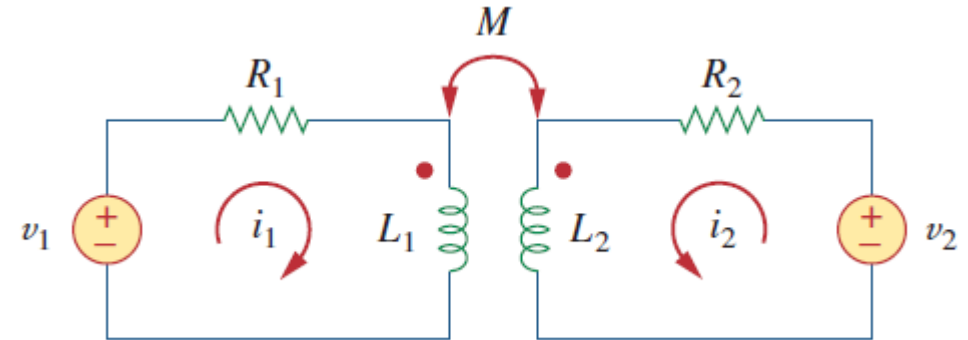
$$L = L_1 + L_2 - 2M \quad (\text{Series-opposing connection})$$

Analysis of Circuits involving Mutual Inductance

- In time domain, for coil1 and coil2, KVL gives

$$v_1 = i_1 R_1 + L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = i_2 R_2 + L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$



- In frequency domain

$$\mathbf{V}_1 = (R_1 + j\omega L_1)\mathbf{I}_1 + j\omega M\mathbf{I}_2$$

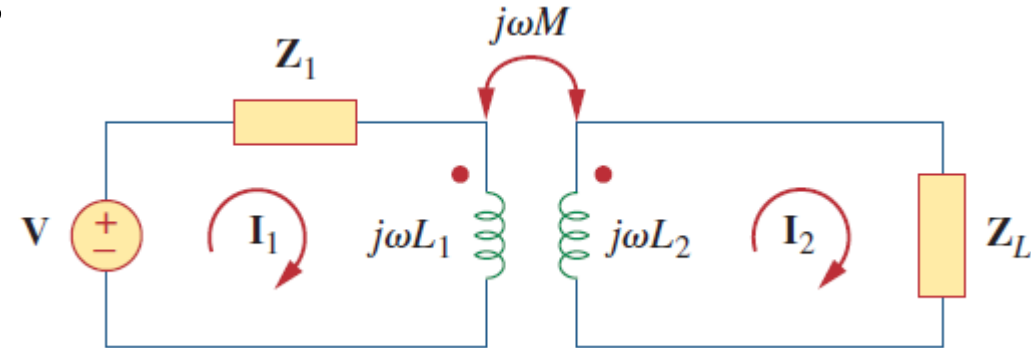
$$\mathbf{V}_2 = j\omega M\mathbf{I}_1 + (R_2 + j\omega L_2)\mathbf{I}_2$$

Analysis of Circuits involving Mutual Inductance

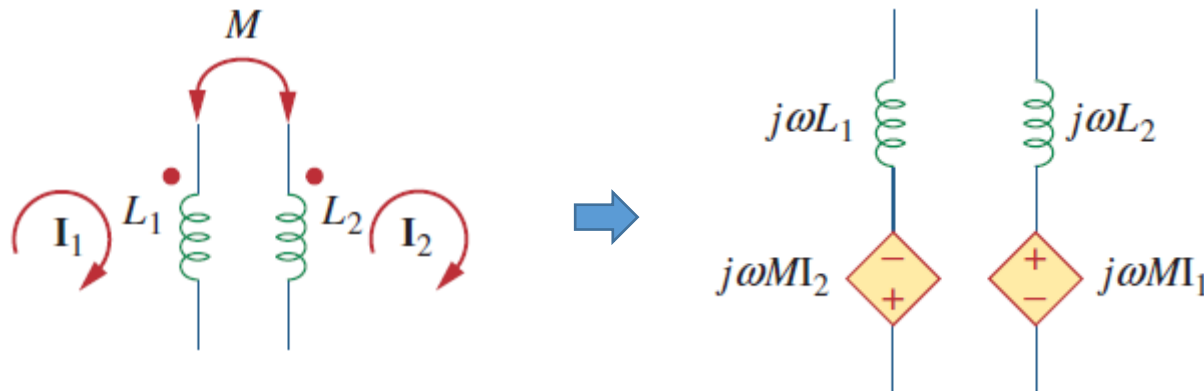
- As a second example
- In frequency domain, for coil1 and coil2, KVL gives

$$V = (Z_1 + j\omega L_1)I_1 - j\omega M I_2$$

$$0 = -j\omega M I_1 + (Z_L + j\omega L_2)I_2$$



- Equivalents with dependent sources

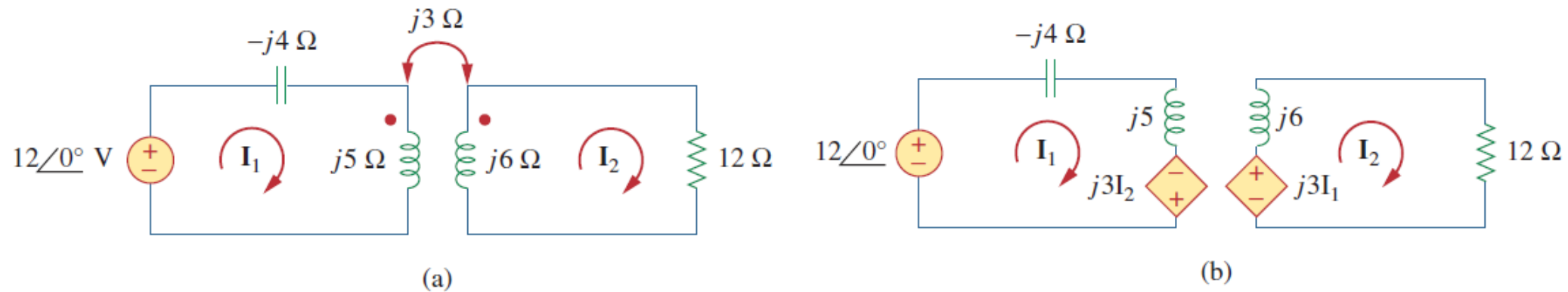


- Note: Please always give a second thought on the signs and the values of the mutually induced voltages

Example: Mutual Inductance

Calculate the phasor currents $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 13.9.

Example 13.1



Solution:

For loop 1, KVL gives

$$-12 + (-j4 + j5)\mathbf{I}_1 - j3\mathbf{I}_2 = 0$$

For loop 2, KVL gives

$$-j3\mathbf{I}_1 + (12 + j6)\mathbf{I}_2 = 0$$



$$\mathbf{I}_2 = \frac{12}{4 - j} = 2.91 \angle 14.04^\circ \text{ A}$$

$$\begin{aligned} \mathbf{I}_1 &= (2 - j4)\mathbf{I}_2 = (4.472 \angle -63.43^\circ)(2.91 \angle 14.04^\circ) \\ &= 13.01 \angle -49.39^\circ \text{ A} \end{aligned}$$

Energy Stored in a Coupled Circuit

- For coupled inductors, the total energy stored depends on the individual inductance and on the mutual inductance

$$w = \frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 \pm Mi_1i_2$$

- The energy stored in the circuit cannot be negative because the circuit is passive

$$\frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 - Mi_1i_2 \geq 0$$

$$\frac{1}{2}(i_1\sqrt{L_1} - i_2\sqrt{L_2})^2 + i_1i_2(\sqrt{L_1L_2} - M) \geq 0$$

$$\sqrt{L_1L_2} - M \geq 0 \quad M \leq \sqrt{L_1L_2}$$

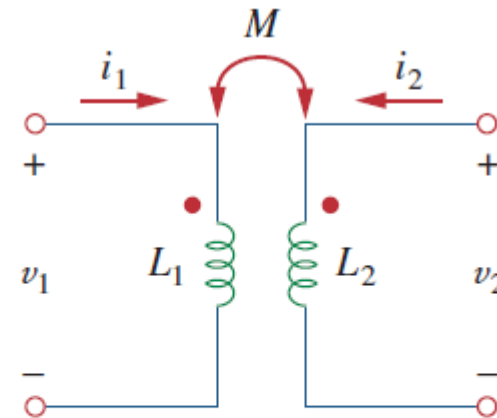


Figure 13.14

The circuit for deriving energy stored in a coupled circuit.

Coefficient of Coupling

- The mutual inductance cannot be greater than the geometric mean of the self-inductances of the coils

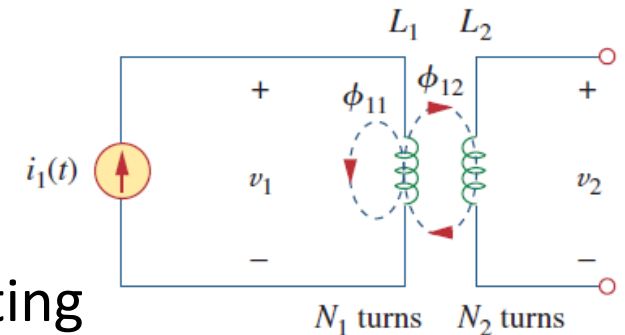
$$M \leq \sqrt{L_1 L_2}$$

- The extent to which the mutual inductance M approaches the upper limit is specified by the **coefficient of coupling** k (耦合系数), given by

$$k = \frac{M}{\sqrt{L_1 L_2}} \quad M = k \sqrt{L_1 L_2}$$

- The coupling coefficient is the fraction of the total flux emanating from one coil that links the other coil. It can be represented

$$k = \frac{\phi_{12}}{\phi_1} = \frac{\phi_{12}}{\phi_{11} + \phi_{12}}$$



The **coupling coefficient** k is a measure of the magnetic coupling between two coils; $0 \leq k \leq 1$.

Example: Coefficient of Coupling (I)

Consider the circuit in Fig. 13.16. Determine the coupling coefficient. Calculate the energy stored in the coupled inductors at time $t = 1$ s if $v = 60 \cos(4t + 30^\circ)$ V.

Solution:

The coupling coefficient is

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{2.5}{\sqrt{20}} = 0.56$$

indicating that the inductors are tightly coupled.

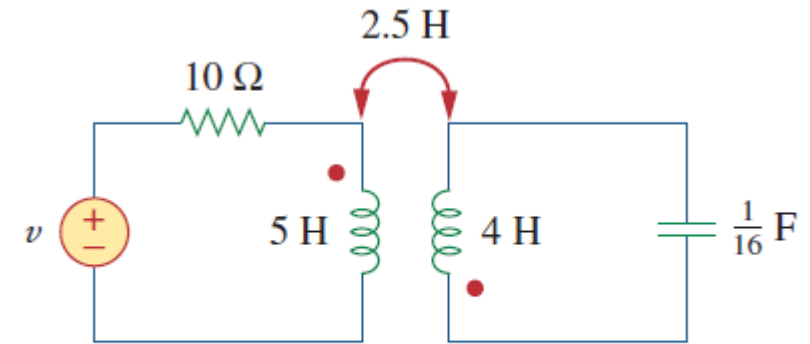
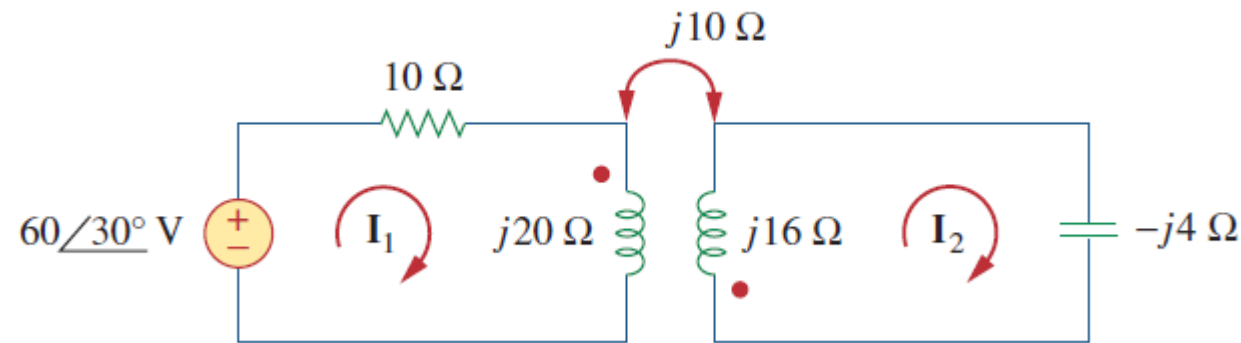
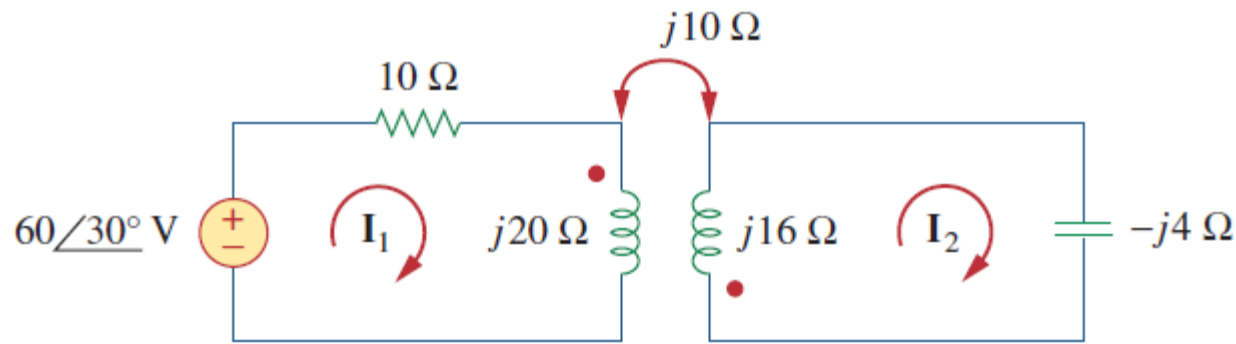


Figure 13.16
For Example 13.3.



The Frequency-Domain Equivalent

Example: Coefficient of Coupling (II)



The Frequency-Domain Equivalent

In the time-domain,

$$i_1 = 3.905 \cos(4t - 19.4^\circ), \quad i_2 = 3.254 \cos(4t + 160.6^\circ)$$

At time $t = 1$ s, $4t = 4$ rad = 229.2° , and

$$i_1 = 3.905 \cos(229.2^\circ - 19.4^\circ) = -3.389 \text{ A}$$

$$i_2 = 3.254 \cos(229.2^\circ + 160.6^\circ) = 2.824 \text{ A}$$

For mesh 1,

$$(10 + j20)\mathbf{I}_1 + j10\mathbf{I}_2 = 60\angle 30^\circ$$

For mesh 2,

$$j10\mathbf{I}_1 + (j16 - j4)\mathbf{I}_2 = 0$$

$$\mathbf{I}_2 = 3.254\angle 160.6^\circ \text{ A}$$



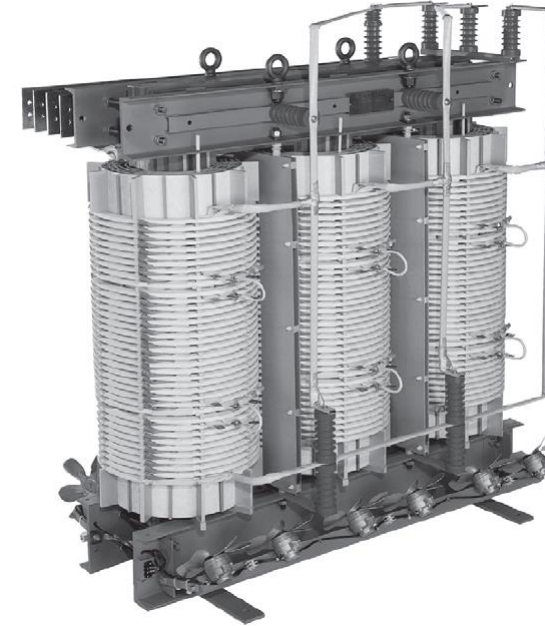
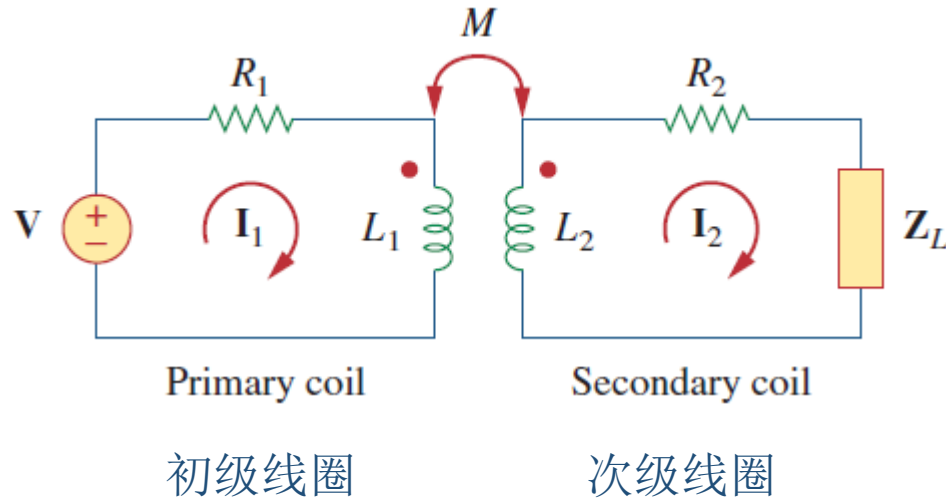
$$\mathbf{I}_1 = -1.2\mathbf{I}_2 = 3.905\angle -19.4^\circ \text{ A}$$

The total energy stored in the coupled inductors is

$$\begin{aligned} w &= \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 + M i_1 i_2 \\ &= 20.73 \text{ J} \end{aligned}$$

Transformers

- A **transformer** (变压器) is a four-terminal device comprising two (or more) magnetically coupled coils



- Applications
 - Step-up or step-down voltage (energy transfer)
 - Isolation coupling
 - Impedance matching (signal transfer)



Linear Transformers

- **Linear Transformers (线性变压器)**: if the coils are wound on a magnetically linear material, viz. a material for which the magnetic permeability is constant
- Linear transformers are sometimes called **air-core transformers (空芯变压器)**, although not all of them are necessary air-core
- Apply KVL to the two loops

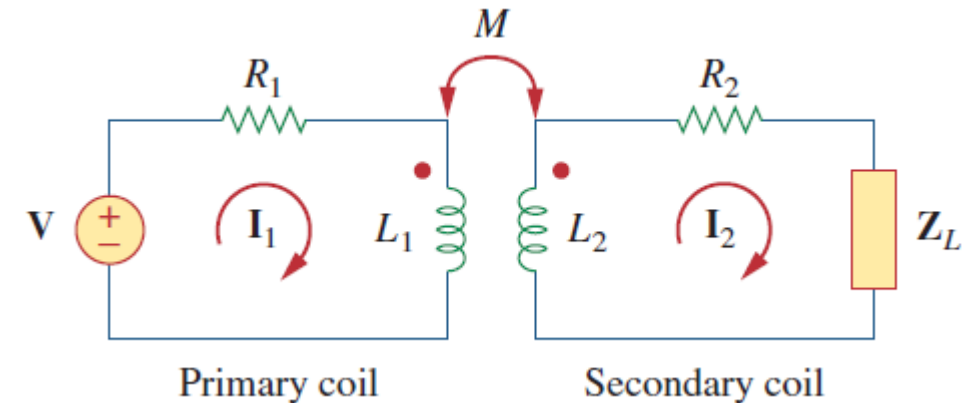
$$\begin{aligned} V &= (R_1 + j\omega L_1)I_1 - j\omega M I_2 \\ 0 &= -j\omega M I_1 + (R_2 + j\omega L_2 + Z_L)I_2 \end{aligned}$$

- Thus the input impedance

$$Z_{in} = \frac{V}{I_1} = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$$

- **Reflected impedance (反射阻抗) or coupled impedance (耦合阻抗)**

$$Z_R = \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$$



Linear Transformers: Equivalent Circuits

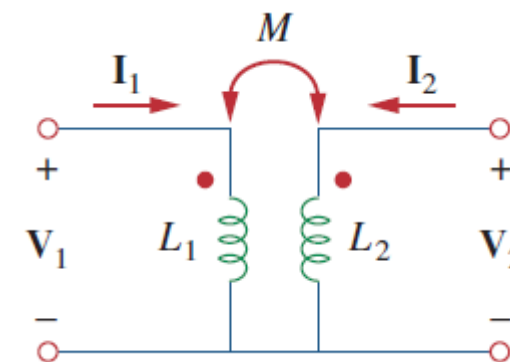
- The coupled inductors can be tricky to work with. One approach is to use a transformation to create an equivalent circuit. The goal is to remove the mutual inductance. This can be accomplished by using a T or a π network
- The goal is to match the terminal voltages and currents from the original network to the new network
- Assume the frequency is ω , apply KVL law to the input and output loops, we have

$$\mathbf{V}_1 = j\omega L_1 \mathbf{I}_1 + j\omega M \mathbf{I}_2$$

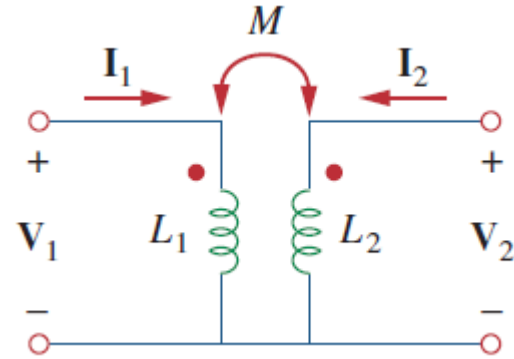
$$\mathbf{V}_2 = j\omega M \mathbf{I}_1 + j\omega L_2 \mathbf{I}_2$$

- Hence in matrix form, that is

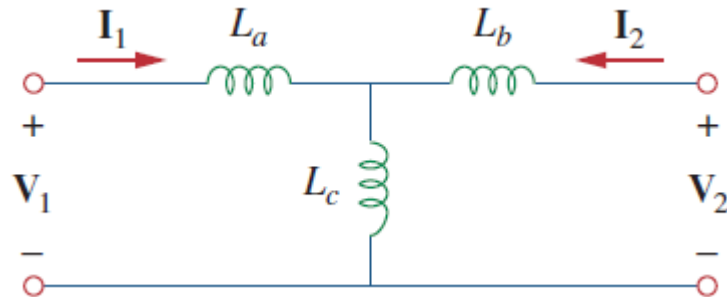
$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$



Equivalent Circuits of Transformer



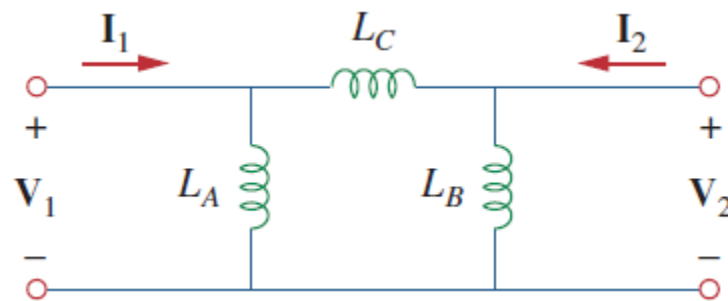
$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$



T-Network Equivalent

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} j\omega(L_a + L_c) & j\omega L_c \\ j\omega L_c & j\omega(L_b + L_c) \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

$$L_a = L_1 - M, \quad L_b = L_2 - M, \quad L_c = M$$



π -Network Equivalent

$$L_A = \frac{L_1 L_2 - M^2}{L_2 - M}, \quad L_B = \frac{L_1 L_2 - M^2}{L_1 - M}$$

$$L_C = \frac{L_1 L_2 - M^2}{M}$$

Example1: Linear Transformers

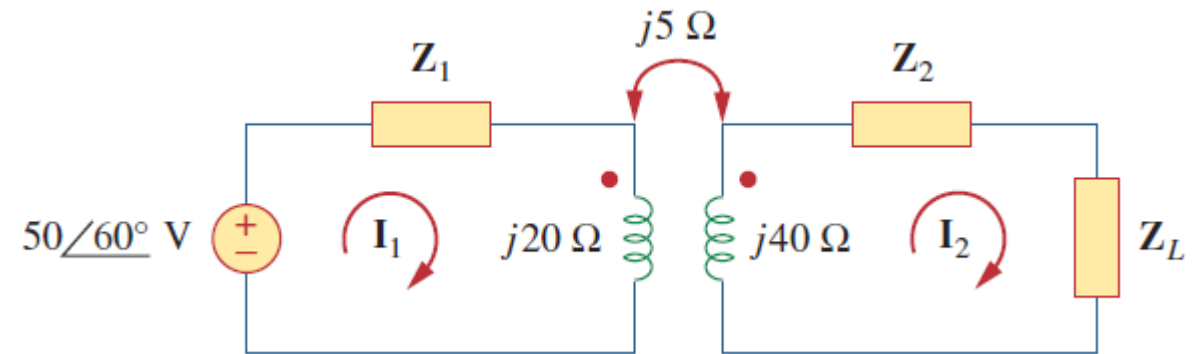
- Q: Calculate the input impedance and current $\mathbf{I}_1$. Take $Z_1 = 60 - j100 \Omega$, $Z_2 = 30 + j40 \Omega$, and $Z_L = 80 + j60 \Omega$

Solution:

$$\begin{aligned}
 \mathbf{Z}_{\text{in}} &= \mathbf{Z}_1 + j20 + \frac{(5)^2}{j40 + \mathbf{Z}_2 + \mathbf{Z}_L} \\
 &= 60 - j100 + j20 + \frac{25}{110 + j140} \\
 &= 60 - j80 + 0.14 \angle -51.84^\circ \\
 &= 60.09 - j80.11 = 100.14 \angle -53.1^\circ \Omega
 \end{aligned}$$

Thus,

$$\mathbf{I}_1 = \frac{\mathbf{V}}{\mathbf{Z}_{\text{in}}} = \frac{50 \angle 60^\circ}{100.14 \angle -53.1^\circ} = 0.5 \angle 113.1^\circ \text{ A}$$



$$\mathbf{V} = (R_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2$$

$$0 = -j\omega M\mathbf{I}_1 + (R_2 + j\omega L_2 + \mathbf{Z}_L)\mathbf{I}_2$$

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{V}}{\mathbf{I}_1} = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L}$$

Example2: Linear Transformers

- Q: Determine the T-equivalent circuit of the linear transformer

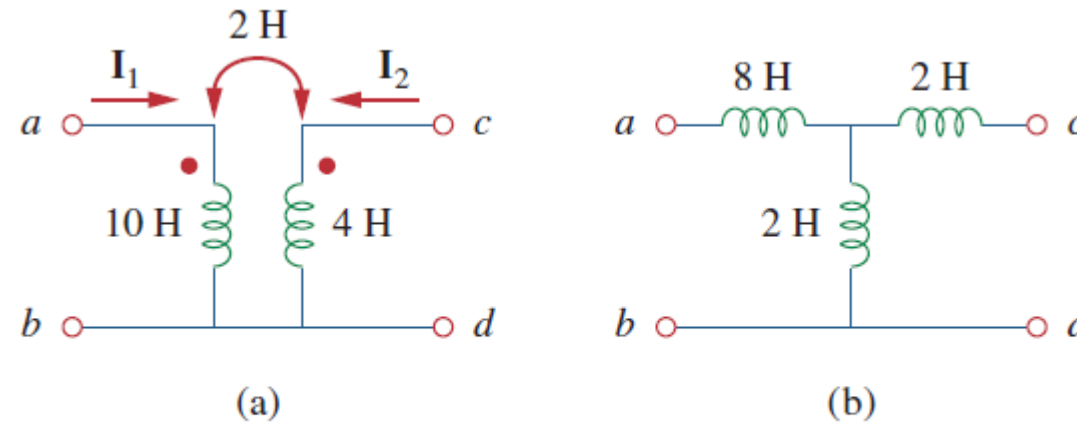


Figure 13.26

For Example 13.5: (a) a linear transformer, (b) its T-equivalent circuit.

Solution:

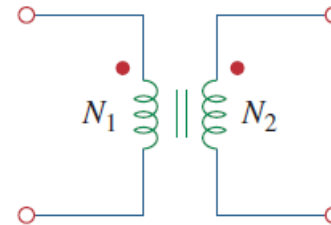
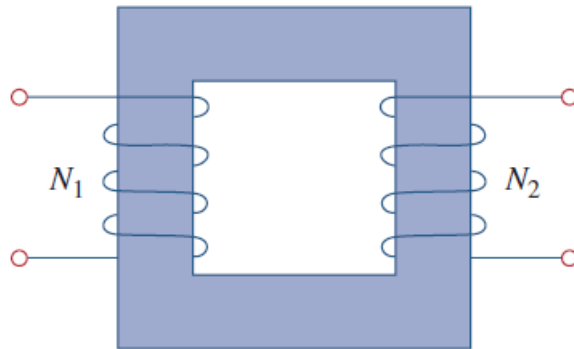
Given that $L_1 = 10$, $L_2 = 4$, and $M = 2$, the T-equivalent network has the following parameters:

$$L_a = L_1 - M = 10 - 2 = 8 \text{ H}$$

$$L_b = L_2 - M = 4 - 2 = 2 \text{ H}, \quad L_c = M = 2 \text{ H}$$

Ideal Transformers

- An **ideal transformer** (理想变压器) is a unity-coupled, lossless transformer in which the primary and secondary coils have infinite self-inductances
 - Perfect coupling ($k = 1$)
 - Coils with very large reactance ($L_1, L_2, M \rightarrow \infty$)



$$\begin{aligned} \mathbf{V}_1 &= j\omega L_1 \mathbf{I}_1 + j\omega M \mathbf{I}_2 \\ \mathbf{V}_2 &= j\omega M \mathbf{I}_1 + j\omega L_2 \mathbf{I}_2 \end{aligned}$$

- The secondary voltage $\mathbf{V}_2 = j\omega L_2 \mathbf{I}_2 + \frac{M \mathbf{V}_1}{L_1} - \frac{j\omega M^2 \mathbf{I}_2}{L_1}$
- With perfect coupling $\mathbf{V}_2 = j\omega L_2 \mathbf{I}_2 + \frac{\sqrt{L_1 L_2} \mathbf{V}_1}{L_1} - \frac{j\omega L_1 L_2 \mathbf{I}_2}{L_1} = \sqrt{\frac{L_2}{L_1}} \mathbf{V}_1 = n \mathbf{V}_1$

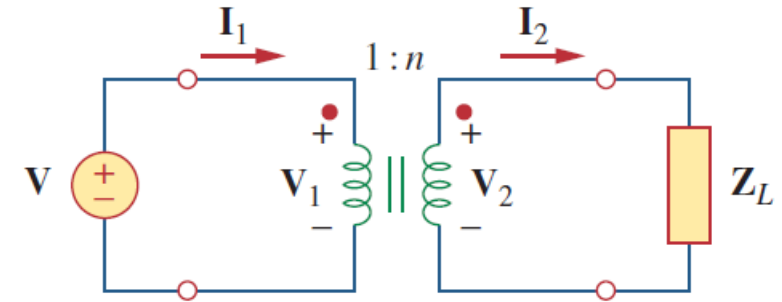
➔ $\frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{N_2}{N_1} = n$

Characteristics of Ideal Transformers (I)

- The voltages are related to each other by the turns ratio n

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} = n$$

- $n > 1$, step-up transformer (升压变压器)
- $n = 1$, isolation transformer (隔离变压器)
- $n < 1$, step-down transformer (降压变压器)



- For the reason of power conservation, the energy supplied to the primary must equal the energy absorbed by the secondary

$$v_1 i_1 = v_2 i_2 \quad \frac{I_1}{I_2} = \frac{V_2}{V_1} = n$$

- The current is related as

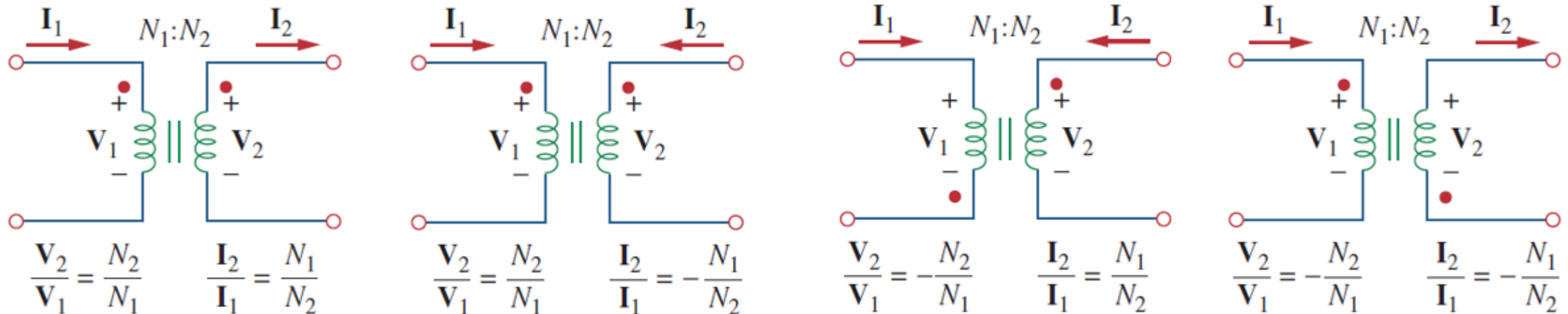
$$\frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{n}$$

Characteristics of Ideal Transformers (II)

- Reflected impedance (the equivalent impedance of load seen from the primary side)

$$Z_{in} = \frac{V_1}{I_1} = \frac{1}{n^2} \frac{V_2}{I_2}$$

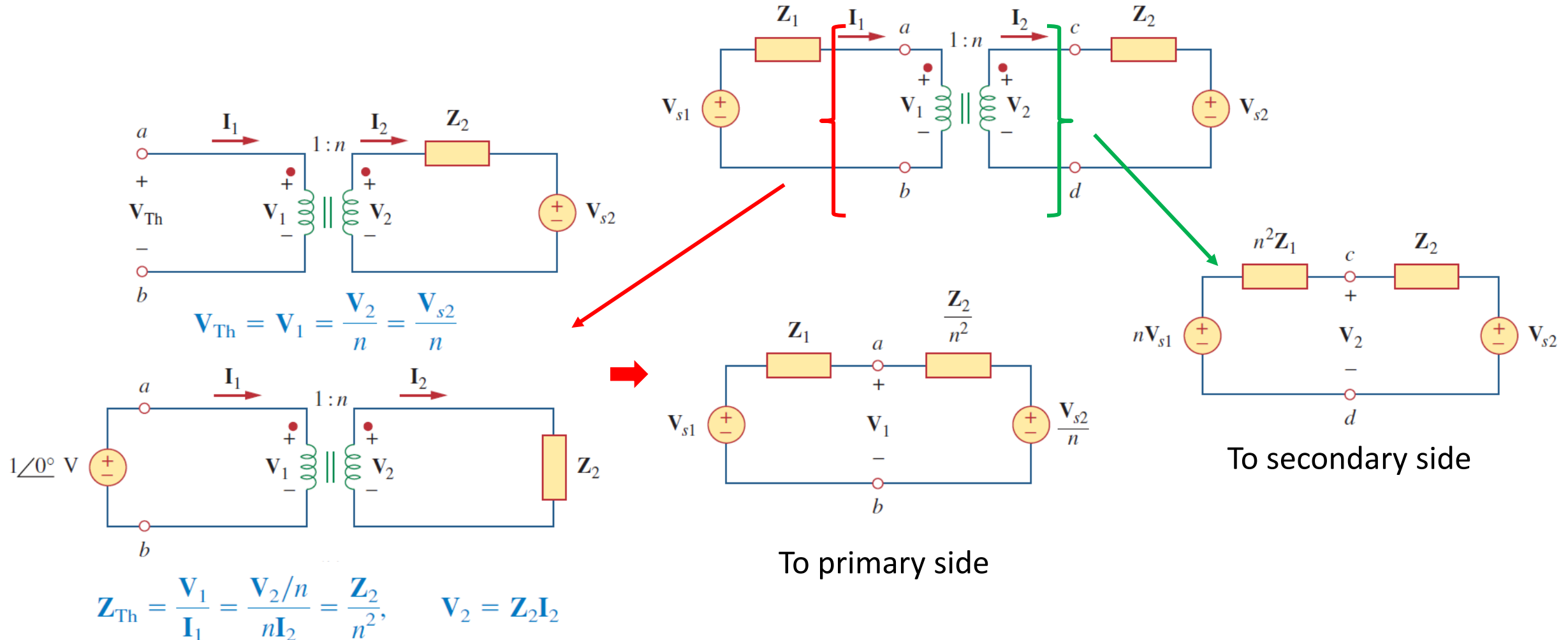
$$V_2/I_2 = Z_L \quad \Rightarrow \quad Z_{in} = \frac{Z_L}{n^2}$$



- If V_1 and V_2 are both positive or both negative at the dotted terminals $\rightarrow +n$ $\frac{V_2}{V_1} = \frac{N_2}{N_1} = n$
- If I_1 and I_2 both enter into or both leave the dotted terminals $\rightarrow -n$ $\frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{n}$

Equivalent Circuits

- Determine the Thevenin equivalent circuit seeing into port a-b or c-d.



Example: Ideal Transformer

- Q: An ideal transformer is rated at 2400/120 V and has 50 turns on the secondary side. Calculate: a) the turns ratio, b) the number of turns on the primary side

Solution:

- a) This is a step-down transformer, since $V_1 = 2400 \text{ V} > V_2 = 120 \text{ V}$

$$n = \frac{V_2}{V_1} = \frac{120}{2,400} = 0.05$$

- b) $N_1 = N_2 / n = 50 / 0.05 = 1000$ turns

Summary

- Two coils are said to be mutually coupled if the magnetic flux emanating from one passes through the other. The mutual inductance between the two coils is given by $M = k \cdot \sqrt{L_1 L_2}$, where k is the coupling coefficient, $0 < k < 1$
- The polarity of the mutually-induced voltage is expressed in the schematic by the dot convention
- The energy stored in two coupled coils is $0.5L_1 i_1^2 + 0.5L_2 i_2^2 \pm M i_1 i_2$
- A transformer is a 4-terminal device containing two or more magnetically coupled coils
It is used in changing the current, voltage, or impedance level in a circuit
- A linear transformer can be replaced by an equivalent T or π network to ease analysis
- An ideal transformer is lossless, unity coupling, and infinite inductances
- Transformer is used as isolation devices and impedance-matching devices for electronics